

	Case Name: Irrigation Canal Repair	Sector	Construction	
	Project Owner: Eder Omar Vilca Carrillo			
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources	
			Schedule with costs	
Processed by	Zahra Hussaini	Risk Analysis	Random simulation	
Date	2025		One of nine std. scenarios	
File Name	C2025-15	Project Control	User defined distributions	
			Automatic tracking	
			Tracking based on user input	

1. Project description

Project authenticity

This project describes the various activities and their interrelations in the rehabilitation and improvement of the Tumanga irrigation canal, located in the locality of Tumanga, Anra District, Huari Province, Ancash Department. The work includes provisional setups, preliminary tasks, reinforced concrete structures, open and covered canal sections, COVID-19 safety measures, environmental mitigation, and quality control actions aimed at ensuring efficient water flow and infrastructure durability. The project consists of activity and cost data that were obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General		Network topology	
# Activities	50	Serial/Parallel (SP)	18%
Planned Duration (PD)	353 days	Activity Distribution (AD)	36%
Budget At Completion (BAC)	268769.55 €	Length of Arcs (LA)	0%
Renewable Resources	8 Types	Topological Float (TF)	38%
Consumable Resources	0 Types		

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	11.6	11.3	2.3
CRI-rho	11.6	11.4	2.4
CRI-tau	10.4	7.6	1.4

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	3.7	12.5	4.0
CRI-rho	3.7	12.5	4.0
CRI-tau	3.7	12.5	4.0

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	15.9	36.3	1.9
SI	20.6	33.6	1.9
SSI	4.1	13.3	3.7
CRI-r	9.8	13.1	2.9
CRI-rho	9.6	13.1	2.9
CRI-tau	13.7	8.4	1.1

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy			Simulated EAC accuracy		
method - PF	MAPE [%]	MPE [%]	method (PF)	MAPE [%]	MPE [%]
PV - 1	125.8	-54.7	1	0	0
PV - SPI	98.2	69.2	CPI	0.1	0
PV - SCI	98.3	69.1	SPI	10.3	10.3
ED - 1	36459.5	36349.8	SPI(t)	9.8	9.8
ED - SPI	98.2	69.2	SCI	10.3	10.3
ED - SCI	98.2	69.1	SCI(t)	9.8	9.8
ES - 1	42.5	-37.6	0.8 CPI + 0.2 SPI	10.2	10.2
ES - SPI(t)	27.4	11.1	0.8 CPI + 0.2 SPI(t)	8.2	8.2
ES - SCI(t)	27.4	11.1			

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 2 tracking periods. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	0	37070	-923.5	1	1.2	0.8	0.9
std dev	0	52424.9	1465.8	0	0.2	0.3	0.02
final	0	0	-1960	1	1	0.5	1

2.3.3.2. Time forecasting

PD	353 days
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Real Duration	363 days
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Late	3%
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EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	303.5	68.6	8.2	8.2
PV - SPI	301	72.1	8.5	8.5
PV - SCI	301	72.1	8.5	8.5
ED - 1	406.5	269.4	6.0	-6.0
ED - SPI	423.5	245.4	8.3	-8.3
ED - SCI	423.5	245.4	8.3	-8.3
ES - 1	467.5	183.1	14.4	-14.4
ES - SPI(t)	467.5	183.1	14.4	-14.4
ES - SCI(t)	467.5	183.1	14.4	-14.4

2.3.3.3. Cost forecasting

BAC	268769.55 €
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Real Cost	268769.55 €
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On budget	0%
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EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	268769.5	0.0	2E-05	2E-05
CPI	268769.5	0.0	2E-05	2E-05
SPI	266986.3	2521.8	0.7	0.7
SPI(t)	268520.0	352.9	0.1	0.1
SCI	266986.3	2521.8	0.7	0.7
SCI(t)	268520.0	352.9	0.1	0.1
0.8 CPI + 0.2 SPI	268305.6	656.1	0.2	0.2
0.8 CPI + 0.2 SPI(t)	268717.9	72.9	0.0	0.0